



JUIN 2024

Bio Source Farm

BUSINESS PLAN



Designed for
**Technical and financial
partners**

Presented by :
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ABOUT US

Company description



LEGAL NAME	BIO SOURCE FARM	LOCALISATION	C5 Ambalamanasy, Toamasina 502
STATUS (On registration)	SOLE PROPRIETORSHIP	NIF/ STAT	6005477481 74908 31 2021 0003 98
LEGAL CREATION	March 2021	ACTIVITY START	June 2023

CORE ACTIVITIES

From 2024 to 2027, BSF plans to boost production, improve infrastructure, and optimize raw material management with a goal of 20% annual growth, while ensuring structuration, quality and innovation.

- **Waste Collection & Valorization :**

BSF takes and converts organic waste from multi-sources into nutrient-rich compost and products for agriculture, promoting a circular economy.

- **Organic Fertilizers :**

Bio Source Frass, a premium organic fertilizer, enhances soil health for various crops, supporting sustainable agriculture.

- **Protein Products :**

We offer protein-rich powders and insect-based pet food from Black Soldier Fly larvae, providing eco-friendly protein alternatives.

- **Training & Capacity Building :**

BSF conducts training programs to empower farmers in adopting sustainable agricultural practices.

- **Innovation & Research :**

Collaborations with research institutions focus on developing resilient crops and improving farming inputs.

TURNING WASTE TO TREASURES

Bio Source Farm (BSF) is a forward-thinking agribusiness startup committed to revolutionizing waste management and agricultural sustainability in Madagascar. Established as a solution to organic waste challenges, BSF specializes in bioconversion through Black Soldier Fly (BSF) larvae, creating high-quality organic products that meet the demands of modern, sustainable farming practices.

TRACTIONS

- *92+ tons of Bio Source Frass sold in the past year, with clients reporting crop yield improvements of up to 30%*
- *Partnerships established with 150+ certified organic producers, including members of SYMABIO, ensuring a stable and expanding customer base.*
- *Development of proprietary larvae-based biostimulants for high-value crops, expanding the product portfolio.*

Executive Summary

Turning waste into treasures



MISSION

To promote high-quality inputs for eco-friendly farming using Black Soldier Fly



VISION

To achieve sustainable Agriculture in Madagascar through waste bioconversion

The Product

Bio Source Frass is the organic fertilizer developed by Bio Source Farm, resulting from the bioconversion of organic waste through Black Soldier Fly (BSF) larvae. BSF also produces protein powders, dried larvae, and pet food, they are **Exofeeds** aimed at promoting sustainable agriculture and circular economy principles.

The Leadership

RASOLOMAMPIONONA Tendry Ny Aina, RASOAVINJATO Mirana, and Mampionona founded Bio Source Farm after collaborating on sustainable agriculture initiatives within Groupe Faly Export. They serve as CEO, CHRO, and COO, respectively.

The Overall Industry

The agricultural inputs industry includes traditional composting methods, chemical fertilizers and feeds. However, Bio Source Farm provides an innovative, eco-friendly alternative by transforming organic waste into high-quality bioconversion products, tailored for sustainable agriculture.

The Competitors

Key competitors for Bio Source Farm include local BSF Farmers, producers of organic fertilizers and waste treatment industry. Bio Source Farm differentiates itself by focusing on a high-nutrient organic product, specifically tailored for various high-value crops, and aiming at niche markets like biological farmers.

The Financial Status

Bio Source Farm needs to raise \$300,000 to begin its extension operations. The company has met 10% of its target thus far. Financial resources will be focused on developing products and sustainable farming education.

Future Plans

By 2025, the company intends to collect 4,500 Tons of food waste before it divert in landfills. By 2030, it expects to reach 1 million farmers in Madagascar and across multiple customers worldwide who use sustainable inputs for Agriculture (crops and animals).



The Organization

An inclusive team motivated by sustainability



Ny Aina Tendry
Chief Executive Officer

Environmental science specialist with a strong background in sustainable agriculture and environmental management. Technical and Process conception



Mirana Carol
Human Resources Officer

Manager graduate and oversees administration and human resources management, fostering a supportive work environment for employee development.



Mampionona
Chief Operations Officer

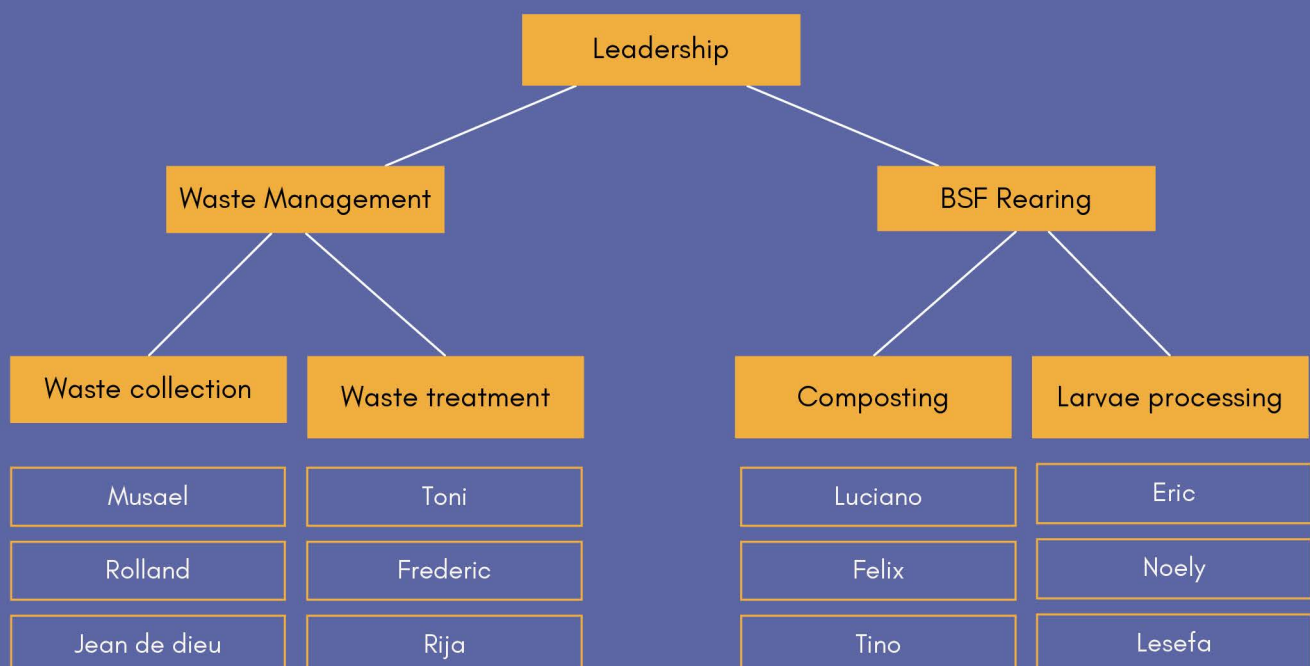
Responsible for production and technical monitoring, ensuring quality control and efficient production processes.



Faly
Senior sales and Counselor

Responsible for external relations and partnerships, he is especially expert in international markets and marketing counselling.

MAP OF THE ORGANIZATION



Business Description

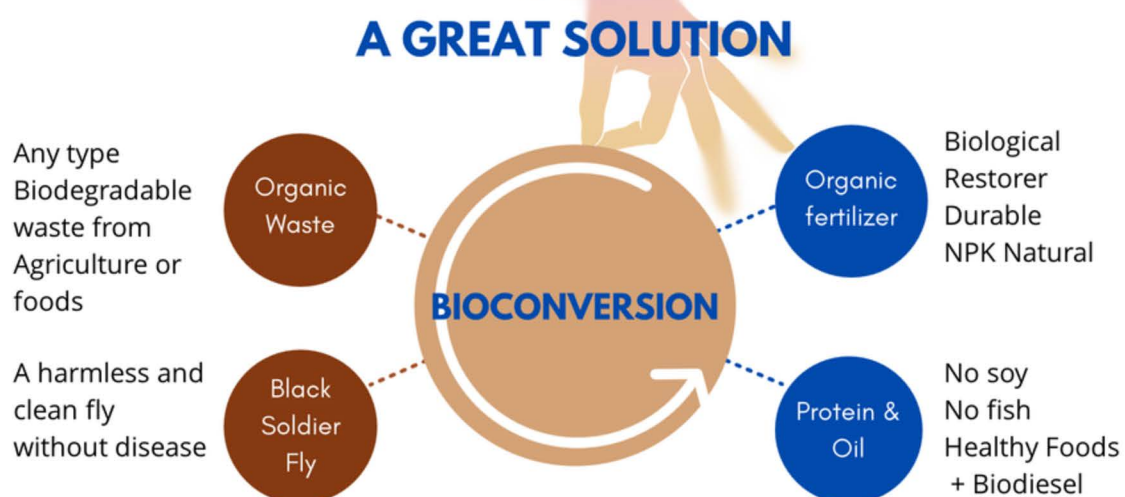
A solution for a right problem in the right time

Problem statement

40% of Madagascar farmland is degraded. It costs annually 1.7 billion USD while threatening food security and environmental health (UNCCD, 2021). Poor management of organic waste adds to this challenge, generating over 3 million tons of waste, yet less than 10% of this is treated (World Bank, 2016). Both create useless land and reduce per capita productivity. Less productivity brings weak human capital and, in the long term, drives persistent poverty. We don't pretend to solve a macroscale problem, but we contribute to the core of its main drivers: soil fertility and agricultural inputs availability to sustain farming and strengthen farmers.

Proposed solution

Bio Source Farm utilizes Black Soldier Fly larvae in an innovative bioconversion process to convert organic waste into nutrient-rich inputs for sustainable agriculture. This method effectively manages waste, offering advantages over traditional composting and chemical fertilizers. It reduces waste volume by up to 80% and produces fertilizers that improve soil health and increase crop yields (*Huang et al.; 2022*). BSF-based fertilizers support long-term soil regeneration (*Zhao et al.; 2024*), unlike chemical options that degrade soil. Additionally, this process helps reduce landfill waste and provides protein alternatives to soymeal and fishmeal, preventing resource overexploitation.



Less space + Less water + Less cost = More products

Objectives

Estimated to be 3 years run at least

MAIN OBJECTIVE

Improve bioconversion capacity of BSFL to process 2,500 Tons of waste annually, and supply 1,500 Tons of Organic fertilizers to support 1,000 local farmers adopting sustainable Agriculture.

Environmental, social, commercial and financial objectives

- Environmental: Divert 2,500 tons of organic waste from landfills annually in Toamasina, reducing greenhouse gas emissions.
- Social: Improve the livelihoods of 1,000 local farmers through access to affordable, eco-friendly inputs (achieving a local-based food self-sufficiency).
- Commercial: Achieve a 30% increase in bioconversion capacity, expanding market share in the organic agriculture sector.
- Financial: Attain 20% revenue growth over the next 12 months.

Expected results, outputs and outcomes

- Environmental: Reduction of 40% in organic waste sent to landfills in Toamasina (Town and peripherals), contributing to a cleaner environment.
- Social: 3,000 smallholder farmers trained in sustainable agriculture practices.
- Commercial: An increase in production by 30%, leading to the supply of 4,500 tons of organic fertilizer.
- Financial: Annual revenue growth of 20%, establishing BSF as a key player in Madagascar's sustainable agriculture market.

KPIs and targets metrics

- Waste diverted from landfills: 2,500 tons annually.
- Farmers supported: 250 farmers trained and supplied with organic fertilizers each 3 months.
- Production capacity: A 30% increase in bioconversion output starting from eggs collection.
- Revenue growth: Achieving 20% annual growth within 12 months.
- Environmental impact: Reduction of waste-related greenhouse gas emissions by 40% in an identified area of concern.

Product List

Our product are produced by treated wastes (Food and Farm waste)



Bio Source Frass **\$0.43/kg**

Organic fertilizer Biostimulant
Exist in 1kg ; 30 kg



Exoboosts **\$0.9/L and \$1.2/L**

Liquid fertilizer and Bioactivators
both obtained from fruits waste
fermentation



Exofeed **\$4/kg - animal feeds**

Protein powder from grinded BSF
larvae fated and soon defatted

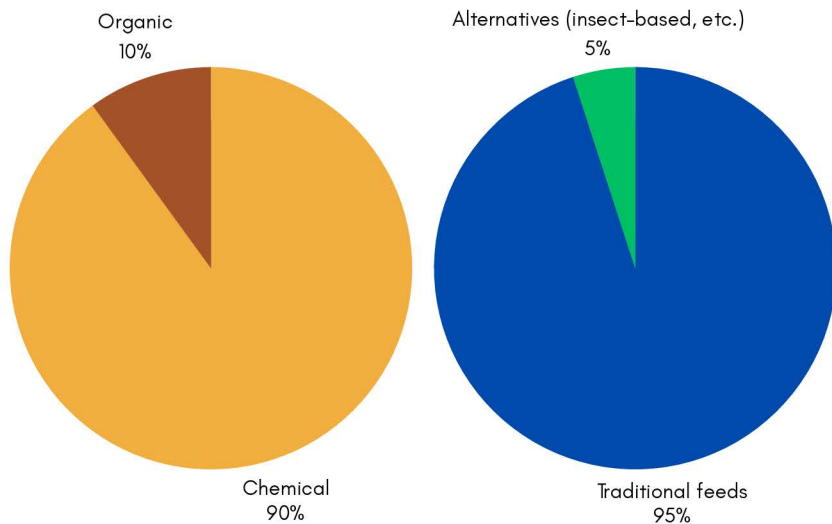


Exofeed **\$3/kg - For pets and fish**

BSF Larvea in a mature stage killed
by pasteurisation and dried in a
oven between 120 - 150 °C

Industry Background

An innovative and rapidly growing industry



Filling A Gap

The big players in the industry are either chemical industry or soy and fishmeal based feeds. This shows that Bio Source Farm addresses the needs of farmers (crops and animals) who aims to develop more sustainable system in a more resilient system of production and consumption.

Source : <https://donnees.banquemondiale.org/pays/madagascar>

THE INDUSTRY

Bio Source Farm addresses the significant gap in **sustainable waste management** and organic **agricultural inputs** within Madagascar's agricultural sector. Traditional farming relies heavily on chemical fertilizers, which can degrade soil quality and harm the environment. Simultaneously, organic waste is often underutilized or improperly disposed of, contributing to pollution.

Bio Source Farm's bioconversion process fills this gap by **turning organic waste** into nutrient-rich organic fertilizers and other bioproducts, providing an environmentally safe, effective solution for soil enrichment and crop health. This approach not only reduces dependency on chemical inputs but also supports a **circular economy** where waste is transformed into valuable resources for local farmers, promoting both environmental and economic sustainability.

Competitor Analysis

What are we doing different?

STRENGTHS

- Convenient and Accessible Products: Eco-friendly fertilizers and insect-based feeds are accessible to farmers, as we are locally present and relatively affordable.
- Innovative Offerings: Unique bioconversion products cater to eco-conscious markets, offering a sustainable alternative to chemical fertilizers.
- Niche Appeal: Growing demand from environmentally aware consumers who prefer organic options.

WEAKNESSES

- Resource-Intensive Production: Bioconversion requires significant resource management and higher costs than traditional fertilizers.
- Supply Chain Limitations: Inconsistent organic waste supply can impact product availability and production scalability.
- Complex Development: Scaling sustainable production across diverse market needs is challenging.

OPPORTUNITIES

- Growing Demand for Sustainable Products: Increasing awareness and demand for organic fertilizers and insect-based feeds in emerging markets.
- Expanding Market Reach: Potential for partnerships with NGOs, government bodies, and agricultural cooperatives to scale impact.
- Technological Advancements: Development of efficient production methods in bioconversion and organic processing.

THREATS

- Competitive Market: New entrants and alternative sustainable agriculture companies add pressure to maintain market position.
- Supply Chain Vulnerabilities: Variability in organic waste supply could affect production consistency.
- Market Education Barriers: Low awareness among farmers on the benefits of organic products may require intensive outreach efforts.

Competitors	Products/Services
BSF Tamatave	BSFL products
Iazafo Compost +	Compost
Taroka	Organic fertilizer
Anjara Env	Compost
Fertiplus	Organic fertilizer
Agrivet	Chemical fertilizer

FEED THE SOIL TO LIVE A FUTURE

The main competitors are in the Atsinanana Region, with some reaching the region through their products. This list focuses on direct competitors based on product similarity and market entry strategies

Market Analysis

THE BIOSTIMULANT TREND

The global organic fertilizer market is projected to reach \$0,5 billion by 2030. In Madagascar, the market for organic agricultural inputs is expanding, with an estimated 10% annual growth rate driven by increasing adoption of sustainable farming practices. Bio Source Farm operates in a niche market for eco-friendly fertilizers derived from bioconversion, with significant potential for scaling as awareness grows. (Future Market Insights Global and Consulting Pvt. Ltd; 2024.)

CUSTOMER PERSONA

Bio source farm products is for a niche customers that are rapidly growing



Beneficiaries

- Farmers: Gain access to affordable, eco-friendly fertilizers that improve soil health and crop yields.
- Local communities: Benefit from reduced organic waste and cleaner environments.
- Youth and women in rural areas: Employment opportunities created through the production and distribution process.

KEY SEGMENTS

- smallholder farmers
- Members of SYMABIO (Syndicat du Bio)
- cooperatives
- Local farmers (Chicken, pig, fish, pets)
- Government

KEY TRENDS

- Increasing demand for organic products .
- Recognition of bad impacts of chemical fertilizers on soil health.
- Promotion sustainable agriculture and organic certifications.

GEOGRAPHIC FOCUS

- Madagascar
- Indian Ocean
- Africa

MARKET SHARES

- 10% till 2028
- 5% till 2030
- 2% till 2030

VALUES

- Life first
- zero waste
- responsible consumption

CUSTOMER NEEDS

- Productivity
- Quality
- Market availability
- Affordability



Our Strategy

Promoting sustainable Agriculture

REACHING A WIDER AUDIENCE

To attract eco-conscious farmers and agriculture businesses, Bio Source Farm aims to emphasize the environmental and economic benefits of using sustainable, organic inputs over traditional options. Our strategy involves educating farmers (including women and vulnerable households) on the long-term soil health benefits of organic fertilizers and the nutritional advantages of insect-based feeds. We plan to leverage partnerships with local agricultural organizations and cooperatives to expand our market reach.

THE 4PS OF MARKETING

Mixing B2B to B2C

PRODUCT

Fertilizers & Proteins

- Purpose: enhance soil fertility and livestock nutrition.
- Customer Needs: Farmers seeking inputs that reduce chemical dependence.
- Unique Selling Points: Biostimulant fertilizers and insect-based feeds

PRICE

Competitive Pricing with Flexible Payment Options (\$0.43 - \$4.36)

- Value Proposition: affordable inputs alternative while emphasizing long-term benefits for soil health.
- Pricing Strategy: discounts for bulk purchases.
- Competitive Positioning: competitive with local and regional organic product suppliers, offering stable average price overtime while pricing trend is high.

PROMOTION

print & broadcast ads, social media, email, search engine, video

- Promotion Channels: Social media, local radio, fairs, and partnerships
- Awareness Tactics: field demonstrations to showcase product effectiveness.
- Key Resources: established partnerships with cooperatives and NGOs

PLACE

physical stores, website, online marketplace

- Availability: through local and online store.
- Store Locations: Focused on regions with high agricultural activity
- Distribution Strategy: Leveraging both in-person distribution points and affiliation.

Changes in climatic parameters, including temperature, humidity, and rainfall, directly influence feedstock availability, productivity and survival of Black Soldier Fly (BSF) at each stage of its lifecycle:

- **Reproduction:** Temperature fluctuations can disrupt BSF mating behavior, reducing the rate of successful egg-laying. Optimal temperatures are critical for adult flies to reproduce effectively. Increased temperatures beyond BSF tolerance levels can result in lower reproduction rates.
- **Egg Development:** Eggs require stable temperature and humidity conditions to develop properly. Extreme heat can lead to dehydration and desiccation of eggs, decreasing hatching success. Conversely, too much moisture can also impact the eggs' viability, making this phase sensitive to both temperature and humidity shifts.
- **Larvae Rearing:** Larvae thrive within a specific temperature and moisture range. Extreme heat can speed up larval metabolism excessively, leading to higher mortality rates. Cold temperatures, on the other hand, slow down growth and reduce waste conversion efficiency, impacting the overall production of organic fertilizer. Inconsistent rainfall patterns and humidity also affect larval health, potentially increasing susceptibility to diseases.
- **Pupation:** The transition from larvae to pupae requires stable environmental conditions. Unpredictable temperature and humidity levels can disrupt this delicate process, resulting in lower survival rates during pupation. High temperatures can accelerate the cycle too quickly, reducing pupal health and leading to deformed or weak adults.

Adaptation

To address these climate-related challenges, Bio Source Farm invest in climate-controlled rearing facilities to stabilize temperature and humidity across all BSF lifecycle stages. **Partnering with 150+ individual and cooperative waste suppliers from diverse regions can also build resilience in the feedstock supply chain, ensuring steady production levels.** Additionally, selecting BSF strains with higher tolerance to temperature fluctuations may improve resilience and productivity under variable climate conditions.

Product efficacy and soil health

Rising temperatures and unpredictable rainfall patterns can influence soil composition, microbial activity, and nutrient cycles, potentially impacting the effectiveness of organic fertilizers. Climate change may increase soil erosion and nutrient leaching, reducing the fertilizers' effectiveness in supporting crop growth.

Adaptation

Bio Source Farm will incorporate soil testing services to tailor fertilizer recommendations according to local climate and soil conditions. We will also research and develop climate-resilient fertilizer formulations that cater to diverse soil needs and help retain moisture, reducing nutrient leaching under varying climate conditions. Finally combine all efforts with farmers and working closely with them to monitor every improvement and crops results.

Changes in Market Demand and Customer Behavior

As climate change affects agricultural productivity, farmers may seek more sustainable, climate-adaptive inputs to improve crop resilience. This shift could increase demand for organic fertilizers and alternative feeds that contribute to sustainable farming practices and reduce dependency on chemical inputs.

Adaptation

Bio Source Farm will leverage this shift by highlighting the role of its products in building climate resilience. Educational marketing that informs farmers about the benefits of organic fertilizers in combating climate-related soil degradation can help strengthen brand loyalty and expand the customer base. We intend to reach over 3,000 farmers using sustainable production system and growing their crops quality by 30% till 2026.

Regulatory and Policy Impacts

Governments are increasingly adopting climate-focused agricultural policies, which may involve restrictions on chemical fertilizer usage or incentives for sustainable farming practices. These policies could either provide growth opportunities for Bio Source Farm or impose new compliance requirements.

Adaptation

We try to work closely with ministries (Agriculture and Industry) and stay updated on regulatory developments and engage with policymakers so we can early align our products with emerging regulations. Offering products that comply with farmers' community consensual needs by integrating cooperatives' platform of agricultural information.

Financial Plan

OUTLOOK FOR CLIMATIC ADAPTATION FUND

Bio Source Farm strategically invests in areas that maximize community impact. By focusing on training, logistics, and farm schools, BSF not only fosters sustainable farming but also supports gender inclusivity and resilience against climate change. This holistic approach ensures that both individuals and the environment benefit from our initiatives.

USE TERMS	VALUE	PERCENTAGE	BENEFICIARIES	INDICATORS
Sustainable farming training	\$22,500	45%	farmers/ cooperatives	1,000 farmers/year
Distribution logistics of fertilizers	\$7,500	15%	local clients around 300 km	500 Tons of BSF distributed
50m ² farm schools	\$20,000	40%	Farmers' wife, children, schools	Increase by 30% of women resilience to climate change
TOTAL	\$50,000	100%		

Bio Source Farm (BSF) seeks for \$50,000 to support sustainable agriculture and community empowerment across three initiatives:

1. Sustainable Farming Training (45%) :
 - Objective : Train 1,000 farmers in sustainable practices and BSF organic fertilizers.
 - Impact : Improved crop yields and soil health.
2. Distribution Logistics of Fertilizers (15%) :
 - Objective : Efficiently deliver 500 tons of BSF fertilizers within 300 km.
 - Impact : Enhanced local farm productivity and reduced chemical fertilizer reliance.
3. 50m² Farm Schools (40%) :
 - Objective : Establish schools for women and children to learn practical farming techniques.
 - Impact : 30% increase in women's resilience to climate change.

Financial Plan

GLOBAL FUNDS NEEDED

Bio Source Farm requires capital to invest in climate-controlled rearing facilities for Black Soldier Fly production, as well as to scale its organic fertilizer and sustainable feed operations. Additional funding is needed for research and development of climate-resilient formulations, marketing, and expanding distribution channels to reach a broader market.

	VALUE	PERCENTAGE
Climate adaptation	\$50,000	17%
Facility expansion	\$80,000	26.5%
Working capital	\$30,000	10%
Research and Development	\$40,000	13%
Market development & Packaging	\$20,000	7%
Machines	\$80,000	26.5%
TOTAL	\$300,000	100%

FINANCIAL OUTLOOK

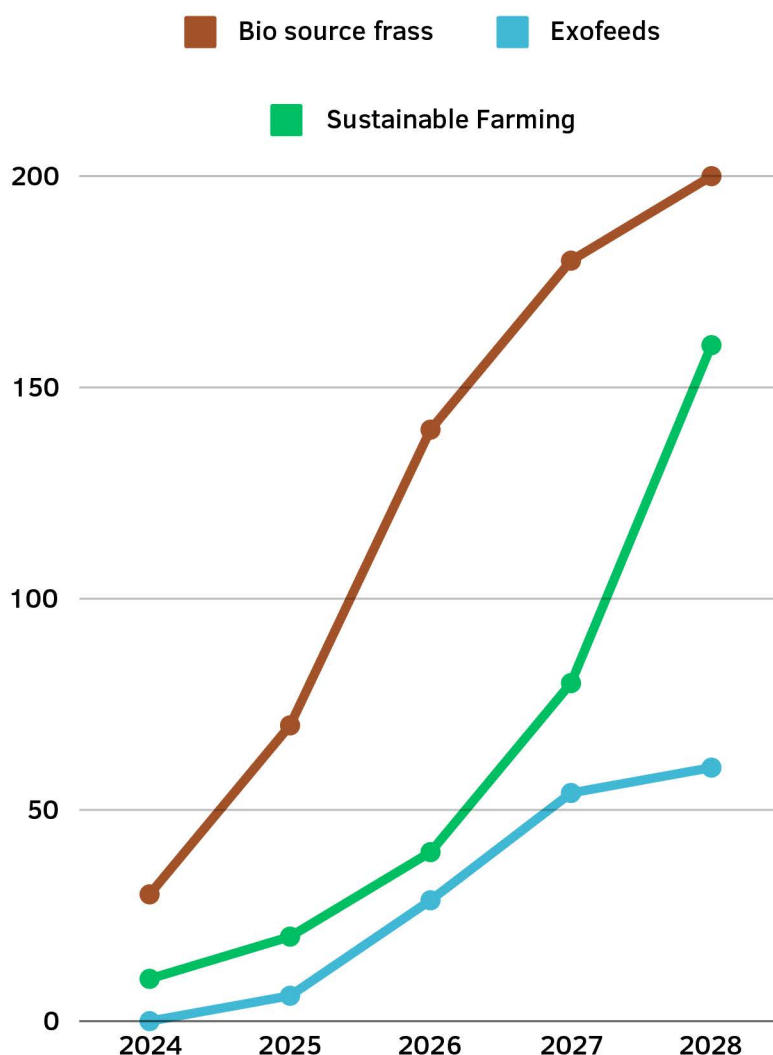
Changing consumer preferences towards sustainable and environmentally friendly options drive high demand for alternative agricultural inputs especially fertilizers and food sources

US\$ M	FY'27	FY'26	FY'25	FY'24	YOY CHANGE
Revenue	0.717	0.287	0.136	0.085	107%
Expenses	0.215	0.136	0.129	0.077	65.5%
Profit	0.502	0.151	0.007	0.008	34.12%
Dividend per share (~10,000 shares)	\$50.2/share	\$15.1/share	\$0.7/share	\$0.8/share	2.47%

Future Plans & Milestones

Evolution of our products over time

- Scale up facilities to increase production capacity for Bio Source Frass (organic fertilizer) and alternative feed products (Exofeeds) derived from Black Soldier Fly (BSF).
- Positions Bio Source Farm to capture more of the growing organic fertilizer market and alternative protein feed industry, especially as demand rises among sustainable and organic agriculture sectors. (10% TAM)
- Enhances Bio Source Farm's R&D capabilities, distribution networks, and influence in the sustainable agriculture sector. Partnerships can also mitigate risks associated with regulatory challenges and costs.



2024

Expand production facilities to increase capacity by 30%



2025

Launch two new products through enhanced R&D efforts.

Launch an educational digital platform for farmer training with 5,000 users.

2026

Implement climate-resilient infrastructure across production sites.

Form at least 3 strategic partnerships for R&D, distribution, and regulatory support.

2027

Establish distribution in two additional African markets.

Achieve profitability with positive cash flow; initial dividend payments.



Contact us for further inquiries



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